

## 1. test cases and intentions

### addcredit (7)

Test folder under tests/ Intention

addcredit/admin\_adds\_to\_other admin adds credit to another user; amount is asked first, then the username

addcredit/exactly\_session\_limit 1000.00 is the highest amount allowed in one session; accepted (boundary)

addcredit/invalid\_amount not a real number; refusd without crashing

addcredit/over\_max\_credit addition would push the account over the 999999.99 maximum; refusd even though it's under the session limit

addcredit/over\_session\_limit 1000.01 is over the session limit; refusd

addcredit/standard\_adds\_own a standard user adds credit to their own account; only the amount is asked

addcredit/unknown\_user admin names a username that doesnt exist; refusd

### bad\_files (4)

Test folder under tests/ intention

bad\_files/corrupt\_accounts\_file accounts file has a bad line; login errors out with a format error

bad\_files/corrupt\_games\_file games file has a bad line; login errors after the username is accepted

bad\_files/missing\_accounts\_file accounts file doesnt exist; login errors immediately

bad\_files/missing\_games\_file games file doesnt exist; login errors after the username is accepted

### buy (8)

Test folder under tests/ intention

buy/admin\_can\_buy confirms an admin account can also buy

buy/already\_owned buying the same game twice in one session; second attempt refusd

buy/buying\_own\_game seller tries to buy their own listing; refusd

buy/game\_not\_found no game with that name exists; refusd right after the name is typed

buy/new\_listing\_same\_session a game listed this same session can't be bought until the next session

buy/not\_enough\_credit buyer's credit is below the game price; refusd

Test folder under tests/ intention

buy/seller\_over\_max\_credit purchase would push the seller over the maximum account credit; refusd

buy/successful\_buy buy-standard user buys a game; checks both questions and the 04 daily line

### create (9)

Test folder under tests/ intention

create/create\_as\_nonadmin full-standard MoonlitKai cant create; refusd before any prompt; not written

create/create\_duplicate\_deleted\_this\_session Nimbus2 deleted then re-created in same session still refusd as already exists; only 02 and 00 written

create/create\_duplicate\_existing existing MoonlitKai refusd as username already exists; user type never asked; not written

create/create\_empty\_username empty new username refusd as invalid username; user type never asked; not written

create/create\_end\_username new username END refusd as invalid username; user type never asked; not written

create/create\_success admin creates NewTester FS; daily gets 01 (zero credit) then 00

create/create\_success\_15char Exactly-15-char FifteenCharUser accepted (boundary); daily 01 then 00

create/create\_too\_long 16-char SixteenCharUser1 refusd as username too long; user type never asked; not written

create/invalid\_user\_type admin create rejects an unsupported account type and writes only the logout record.

### delete (6)

Test folder under tests/ intention

delete/delete\_as\_nonadmin full-standard MoonlitKai cant delete; refusd before any prompt; not written

**delete/delete\_empty\_username delete with empty username refusd as invalid username; not written**

delete/delete\_end\_username delete with END refusd as invalid username; not written

delete/delete\_nonexistent delete of GhostAccount99 refusd as no such user; not written

delete/delete\_self\_admin admin deletes self QuillFox9; session ends at once; daily gets 02 then 00 with no typed logout

delete/delete\_success admin deletes Nimbus2; daily gets 02 with their credit then 00

### general (5)

Test folder under tests/ intention

general/invalid\_empty\_line blank line as transaction while logged in refusd as invalid transaction; not written

general/invalid\_unknown\_word unknown word frobnicate while logged in refusd as invalid transaction; not written

general/invalid\_wrong\_case word login with wrong case refusd as invalid transaction (case-sensitive); not written

general/permission\_buy\_as\_sellstandard sell-standard CraterJax cant buy; refusd before buy prompts; not written

general/permission\_sell\_as\_buystandard buy-standard Thistledown7 cant sell; refusd before sell prompts; not written

#### **list (5)**

Test folder under tests/ intention

list/games\_for\_sale lists every game curently for sale with name, seller, and price

list/nothing\_for\_sale games file has nothing listed; prints the no-games message

list/read\_only running list twice in a row gives identical output; nothing is written to the daily file

list/reflects\_new\_listing a game listed earlier in the session shows up in list right after

list/stays\_after\_purchase a bought game remains listed, since digital copies don't run out

#### **login (8)**

Test folder under tests/ intention

login/login\_case\_mismatch lowercase quillfox9 refusd as no such user (case-sensitive); no daily file

login/login\_empty\_username empty username on login refusd as invalid username; no session so no daily file

login/login\_end\_username username END on login refusd as invalid username; no daily file

login/login\_second\_login\_refused second login while logged in refusd without asking username; only logout writes daily

login/login\_success\_admin admin QuillFox9 can log in and log out; daily gets one 00 line

login/login\_success\_standard full-standard MoonlitKai can log in and log out; daily gets one 00 line

login/login\_unknown\_user unknown GhostAccount99 refusd; next logout refusd becuse never logged in; no daily file

login/transactions\_before\_login create, delete, logout before login each refusd with please login first; only later login writes daily

#### **logout (3)**

Test folder under tests/ intention

logout/logout\_after\_logout logout after a good logout refusd; daily keeps only the first 00 line

logout/logout\_blocks\_until\_login after logout, create refusd; new login starts second session; daily reflects only the newer session, one 00 line

logout/logout\_not\_logged\_in logout with nobody logged in refusd; no daily file

#### **refund (8)**

Test folder under tests/ intention

refund/buyer\_over\_max\_credit refund would push the buyer over the maximum account credit; refusd

refund/invalid\_amount checks negative, zero, and nonnumeric amounts in one session; all refusd

refund/refund\_as\_nonadmin a standard account cant refund; rejection happens before buyer or seller

#### **prompts.**

refund/same\_person\_twice same username entered as buyer and seller; refusd before an amount is asked

refund/seller\_not\_enough\_credit seller's credit is below the refund amount; refusd

refund/successful\_refund admin moves credit from seller back to buyer; checks all three questions and the 05 line

refund/unknown\_person buyer username doesnt exist; refusd before further questions

refund/unknown\_seller refund rejects an unknown seller after accepting an existing buyer.

#### **sell (11)**

Test folder under tests/ intention

sell/admin\_can\_sell confirms an admin account can also sell, not just standard users

sell/invalid\_price price is not a number; refusd without crashing

sell/name\_END END is the file's end marker, so it can't be used as a game name

sell/name\_already\_listed game name already for sale by a diffrent seller; refusd, since uniqueness is checked globally across sellers

sell/name\_exactly\_25 25-character game name accepted (boundary)

sell/name\_too\_long 26-character game name refusd before the price prompt

sell/price\_at\_max price of exactly 999.99 is accepted (boundary)

sell/price\_over\_max price of 1000.00 is over the 999.99 maximum; refusd before anything is saved

sell/sellstandard\_can\_sell confirms the positive case: a sell-standard account can list a game

Test folder under tests/ intention

sell/successful\_sell full-standard user sells a game normally; checks both questions and the 03 daily line

sell/whole\_dollar\_price price typed with no decimal (15) is saved zero-padded as 015.00

## 2. requirements problems and decisions

we used the course handout, phase 1 assignment, lab 1 update and slack answers. the table records our choices where

the requirments were unclear.

issue decision / handling coverage

starting credit we use 0.00 for a new account and do not ask for starting credit.

the create/create\_success; creat

slack answers conflict. we will ask the client about it after the e/create\_success\_15char blackout ends.

File lengths We use the example lengths: username 15, game name 25, credit 9 check\_formats.sh; and price 6. account lines are 28 characters and game lines are 48. name-length tests END lines are padded to the same lengths.

permissions and errors we check permission before asking for transaction details. where no

create/invalid\_user\_type;

exact error was given, we chose the wording used in the test files. an permission

tests; invalid account type gives ERROR: invalid user type. refund/refund\_as\_nonadmin

failed transactions a failed transaction has no daily-file record. if login never succeeds, failure tests and bad\_files

there should be no daily file. tests

multiple sessions our choice is to overwrite the daily file at each logout. after

two logout/logout\_blocks\_until\_lo

sessions, only the second session is in the final file. gin

admin self-delete an admin can delete their own account. we chose to end the

session delete/delete\_self\_admin

straight away, writing the 02 record followed by 00.

game names and list we use unique game names across sellers and choose sell/name\_already\_listed;

list case-sensitive matching. list uses file order and puts new listings at tests

the end. buying a game does not remove its listing. the

duplicate-name test does not separately test case sensitivity.

credit and refunds zero and negative amounts are rejected for refund and addcredit. a

amount, insufficient-credit

refund takes credit from the seller, who must have enough to cover it. and overflow

tests credit cant go above 999999.99.

addcredit limit we chose a cumulative 1000.00 limit per account per session. these

### addcredit/exactly\_session\_li

tests check single amounts at and above the limit. adding several mit;

amounts and resetting the limit still need later testing. addcredit/over\_session\_limit

collection and deleted these input files do not contain the game collection. they cant test not tested by this suite

data stored ownership, removing a refunded game or buying it again after

a refund. we still need to decide how to handle a deleted user's

attached data.

File paths Until the client confirms how filenames are passed, our runner run\_test.sh

supplies accounts.txt, games.txt and the daily-file path as

command-line arguments.

## 3. shared starting fixtures

Normal tests use copies of these accounts.txt and games.txt files. Tests with different or missing files are marked in Section

**4. we print the standard files once rather than repeat them for every test. padding spaces are kept in the records.**

accounts.txt

```
QuillFox9 AA 000500.00
ByteHarbor AA 000000.00
MoonlitKai FS 000200.00
Thistledown7 BS 000100.00
CraterJax SS 000050.00
Nimbus2 FS 000005.00
SilverQuasar1 FS 999950.00
ShadowLynxRider FS 000100.00
END
```

games.txt

```
Starlit Driftwood MoonlitKai 015.00
Emberfall Circuit CraterJax 030.00
Glass Orchard MoonlitKai 025.50
Quiet Nova Protocol SilverQuasar1 100.00
Paperclip Expedition QuillFox9 010.00
Driftwood Signal Nine ShadowLynxRider 050.00
Marrow Tide CraterJax 005.00
END
```

#### 4. test input and expected output

Each heading is a test folder in the repository. input.txt is what the user enters; expected\_terminal.txt is what the program prints. The input is not echoed in the terminal output. If expected\_daily.txt is absent, the program should not create a daily file.

tests/addcredit/admin\_adds\_to\_other  
input.txt expected\_terminal.txt

```
login Enter transaction:
QuillFox9 Enter username:
addcredit Login successful.
50.00 Enter transaction:
MoonlitKai Enter amount of credit:
logout Enter username:
Credit added.
Enter transaction:
Logout successful.
Enter transaction:
```

expected\_daily.txt

```
06 MoonlitKai FS 000050.00
00 QuillFox9 AA 000500.00
```

tests/addcredit/exactly\_session\_limit  
input.txt expected\_terminal.txt

```
login Enter transaction:
QuillFox9 Enter username:
addcredit Login successful.
1000.00 Enter transaction:
MoonlitKai Enter amount of credit:
logout Enter username:
Credit added.
```

```
Enter transaction:
Logout successful.
```

**Enter transaction:**

expected\_daily.txt

```
06 MoonlitKai FS 001000.00
00 QuillFox9 AA 000500.00
```

tests/addcredit/invalid\_amount  
input.txt expected\_terminal.txt

```
login Enter transaction:
QuillFox9 Enter username:
addcredit Login successful.
lots Enter transaction:
addcredit Enter amount of credit:
0.00 ERROR: invalid amount
addcredit Enter transaction:
-1.00 Enter amount of credit:
logout ERROR: invalid amount
Enter transaction:
Enter amount of credit:
ERROR: invalid amount
Enter transaction:
Logout successful.
Enter transaction:
```

```
expected_daily.txt
00 QuillFox9 AA 000500.00

tests/addcredit/over_max_credit
input.txt expected_terminal.txt

login Enter transaction:
QuillFox9 Enter username:
addcredit Login successful.
100.00 Enter transaction:
SilverQuasar1 Enter amount of
credit: logout Enter username:
ERROR: credit limit exceeded
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt
00 QuillFox9 AA 000500.00

tests/addcredit/over_session_limit
input.txt expected_terminal.txt

login Enter transaction:
QuillFox9 Enter username:
addcredit Login successful.
1000.01 Enter transaction:
MoonlitKai Enter amount of credit:
logout Enter username:
ERROR: session credit limit of 1000.00
exceeded Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt
00 QuillFox9 AA 000500.00

tests/addcredit/standard_adds_own
input.txt expected_terminal.txt

login Enter transaction:
MoonlitKai Enter username:
addcredit Login successful.
50.00 Enter transaction:
logout Enter amount of credit:
Credit added.
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt
06 MoonlitKai FS 000050.00
00 MoonlitKai FS 000250.00

tests/addcredit/unknown_user
input.txt expected_terminal.txt

login Enter transaction:
QuillFox9 Enter username:
addcredit Login successful.
10.00 Enter transaction:
Ghostwalker12 Enter amount of credit:
logout Enter username:
ERROR: no such user
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt
00 QuillFox9 AA 000500.00

tests/bad_files/corrupt_accounts_file
accounts.txt differs from the standard fixture and is printed
below. input.txt expected_terminal.txt

login Enter transaction:
QuillFox9 ERROR: user accounts file is not in the correct
format logout Enter transaction:
ERROR: invalid transaction
Enter transaction:
ERROR: please login first
Enter transaction:
```

```
expected_daily.txt: absent. No daily transaction file is
expected. accounts.txt

QuillFox9 AA 000500.00
ByteHarbor AA 000000.00
MoonlitKai FS 0
ThistleDown7 BS 000100.00
CraterJax SS 000050.00
Nimbus2 FS 000005.00
SilverQuasar1 FS 999950.00
ShadowLynxRider FS 000100.00
END
```

```
tests/bad_files/corrupt_games_file
games.txt differs from the standard fixture and is printed
below. input.txt expected_terminal.txt
```

```
login Enter transaction:
QuillFox9 Enter username:
logout ERROR: available games file is not in the correct
format Enter transaction:
ERROR: please login first
Enter transaction:
```

```
expected_daily.txt: absent. No daily transaction file is
expected. games.txt
```

```
Starlit Driftwood MoonlitKai 015.00
Emberfall Circuit Crat
Glass Orchard MoonlitKai 025.50
Quiet Nova Protocol SilverQuasar1 100.00
Paperclip Expedition QuillFox9 010.00
Driftwood Signal Nine ShadowLynxRider 050.00
Marrow Tide CraterJax 005.00
END
```

```
tests/bad_files/missing_accounts_file
accounts.txt is intentionally absent.
```

```
input.txt expected_terminal.txt
```

```
login Enter transaction:
QuillFox9 ERROR: cannot open user accounts file
logout Enter transaction:
ERROR: invalid transaction
Enter transaction:
ERROR: please login first
Enter transaction:
```

```
expected_daily.txt: absent. No daily transaction file is
expected. tests/bad_files/missing_games_file
```

**games.txt is intentionally absent.**

```
input.txt expected_terminal.txt
```

```
login Enter transaction:
QuillFox9 Enter username:
logout ERROR: cannot open available games file
Enter transaction:
ERROR: please login first
Enter transaction:
```

```
expected_daily.txt: absent. No daily transaction file is expected.
```

```
tests/buy/admin_can_buy
input.txt expected_terminal.txt
```

```
login Enter transaction:
QuillFox9 Enter username:
buy Login successful.
Starlit Driftwood Enter transaction:
MoonlitKai Enter game name:
logout Enter seller username:
Game purchased.
Enter transaction:
Logout successful.
Enter transaction:
```

```
expected_daily.txt
```

```
04 Starlit Driftwood MoonlitKai QuillFox9 015.00 00
QuillFox9 AA 000485.00
```

```
tests/buy/already_owned
input.txt expected_terminal.txt
```

```
login Enter transaction:
Thistledown7 Enter username:
buy Login successful.
Starlit Driftwood Enter transaction:
MoonlitKai Enter game name:
buy Enter seller username:

Starlit Driftwood Game purchased.
MoonlitKai Enter transaction:
logout Enter game name:
Enter seller username:
ERROR: game already owned
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt

04 Starlit Driftwood MoonlitKai Thistledown7 015.00 00
Thistledown7 BS 000085.00

tests/buy/buying_own_game
input.txt expected_terminal.txt

login Enter transaction:
MoonlitKai Enter username:
buy Login successful.
Starlit Driftwood Enter transaction:
MoonlitKai Enter game name:
logout Enter seller username:
ERROR: cannot buy your own game
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt

00 MoonlitKai FS 000200.00

tests/buy/game_not_found
input.txt expected_terminal.txt

login Enter transaction:
MoonlitKai Enter username:
buy Login successful.
Nothing Here Enter transaction:
logout Enter game name:
ERROR: game not found
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt

00 MoonlitKai FS 000200.00

tests/buy/new_listing_same_session
input.txt expected_terminal.txt

login Enter transaction:
QuillFox9 Enter username:
sell Login successful.
Fresh Copper Ledger Enter transaction:
12.00 Enter game name:
buy Enter price:
Fresh Copper Ledger Game put up for sale.
QuillFox9 Enter transaction:
logout Enter game name:
Enter seller username:
ERROR: game not available for purchase this
session Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt

03 Fresh Copper Ledger QuillFox9 012.00 00
QuillFox9 AA 000500.00

tests/buy/not_enough_credit
input.txt expected_terminal.txt

login Enter transaction:
Nimbus2 Enter username:
buy Login successful.
Emberfall Circuit Enter transaction:
CraterJax Enter game name:
```

```
logout Enter seller username:
ERROR: not enough credit
Enter transaction:
Logout successful.
Enter transaction:
expected_daily.txt

00 Nimbus2 FS 000005.00

tests/buy/seller_over_max_credit
input.txt expected_terminal.txt

login Enter transaction:
MoonlitKai Enter username:
buy Login successful.
Quiet Nova Protocol Enter transaction:
SilverQuasar1 Enter game name:
logout Enter seller username:
ERROR: seller credit limit exceeded
Enter transaction:
Logout successful.
Enter transaction:
expected_daily.txt

00 MoonlitKai FS 000200.00

tests/buy/successful_buy
input.txt expected_terminal.txt

login Enter transaction:
Thistledown7 Enter username:
buy Login successful.
Starlit Driftwood Enter transaction:
MoonlitKai Enter game name:
logout Enter seller username:
Game purchased.
Enter transaction:
Logout successful.
Enter transaction:
expected_daily.txt

04 Starlit Driftwood MoonlitKai Thistledown7 015.00 00
Thistledown7 BS 000085.00
tests/create/create_as_nonadmin
input.txt expected_terminal.txt

login Enter transaction:
MoonlitKai Enter username:
create Login successful.
logout Enter transaction:
ERROR: not allowed for this account
type Enter transaction:
Logout successful.
Enter transaction:
expected_daily.txt

00 MoonlitKai FS 000200.00

tests/create/create_duplicate_deleted_this_sessio
n input.txt expected_terminal.txt

login Enter transaction:
QuillFox9 Enter username:
delete Login successful.
Nimbus2 Enter transaction:
create Enter username to delete:
Nimbus2 User deleted.
logout Enter transaction:
Enter new username:
ERROR: username already exists
Enter transaction:
Logout successful.
Enter transaction:
expected_daily.txt

02 Nimbus2 FS 000005.00
00 QuillFox9 AA 000500.00

tests/create/create_duplicate_existin
g input.txt expected_terminal.txt

login Enter transaction:
QuillFox9 Enter username:
create Login successful.
```

```
MoonlitKai Enter transaction:
logout Enter new username:
ERROR: username already exists
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt

00 QuillFox9 AA 000500.00

tests/create/create_empty_username
input.txt expected_terminal.txt

login Enter transaction:
QuillFox9 Enter username:
create Login successful.
Enter transaction:
logout Enter new username:
ERROR: invalid username
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt

00 QuillFox9 AA 000500.00

tests/create/create_end_username
input.txt expected_terminal.txt

login Enter transaction:
QuillFox9 Enter username:
create Login successful.
END Enter transaction:
logout Enter new username:
ERROR: invalid username
Enter transaction:
Logout successful.
Enter transaction:
expected_daily.txt

00 QuillFox9 AA 000500.00

tests/create/create_success
input.txt expected_terminal.txt

login Enter transaction:
QuillFox9 Enter username:
create Login successful.
NewTester Enter transaction:
FS Enter new username:
logout Enter user type (AA, FS, BS,
SS): User created.
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt

01 NewTester FS 000000.00
00 QuillFox9 AA 000500.00

tests/create/create_success_15cha
r input.txt expected_terminal.txt

login Enter transaction:
QuillFox9 Enter username:
create Login successful.
FifteenCharUser Enter
transaction: SS Enter new
username:
logout Enter user type (AA, FS, BS,
SS): User created.
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt

01 FifteenCharUser SS
000000.00 00 QuillFox9 AA
000500.00

tests/create/create_too_long
input.txt expected_terminal.txt

login Enter transaction:
```

```
QuillFox9 Enter username:
create Login successful.
SixteenCharUser1 Enter
transaction: logout Enter new
username:
ERROR: username too long (max
15) Enter transaction:
Logout successful.
Enter transaction:
expected_daily.txt

00 QuillFox9 AA 000500.00

tests/create/invalid_user_type
input.txt expected_terminal.txt

login Enter transaction:
QuillFox9 Enter username:
create Login successful.
Wrenfeather3 Enter transaction:
XX Enter new username:
logout Enter user type (AA, FS, BS,
SS): ERROR: invalid user type
Enter transaction:
Logout successful.
Enter transaction:
expected_daily.txt

00 QuillFox9 AA 000500.00

tests/delete/delete_as_nonadmi
n input.txt
expected_terminal.txt

login Enter transaction:
MoonlitKai Enter username:
delete Login successful.
logout Enter transaction:
ERROR: not allowed for this account type
Enter transaction:
Logout successful.
Enter transaction:
expected_daily.txt

00 MoonlitKai FS 000200.00

tests/delete/delete_empty_usernam
e input.txt expected_terminal.txt

login Enter transaction:
QuillFox9 Enter username:
delete Login successful.
Enter transaction:
logout Enter username to
delete: ERROR: invalid username
Enter transaction:
Logout successful.
Enter transaction:
expected_daily.txt

00 QuillFox9 AA 000500.00

tests/delete/delete_end_usernam
e input.txt
expected_terminal.txt

login Enter transaction:
QuillFox9 Enter username:
delete Login successful.
END Enter transaction:
logout Enter username to
delete: ERROR: invalid username
Enter transaction:
Logout successful.
Enter transaction:
expected_daily.txt

00 QuillFox9 AA 000500.00

tests/delete/delete_nonexisten
t input.txt
expected_terminal.txt
```

```
login Enter transaction:
QuillFox9 Enter username:
delete Login successful.
GhostAccount99 Enter transaction:
logout Enter username to delete:
ERROR: no such user
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt

00 QuillFox9 AA 000500.00

tests/delete/delete_self_admin
input.txt
expected_terminal.txt

login Enter transaction:
QuillFox9 Enter username:
delete Login successful.
QuillFox9 Enter
transaction: Enter username
to delete:
User deleted.
Enter transaction:

expected_daily.txt

02 QuillFox9 AA 000500.00
00 QuillFox9 AA 000500.00

tests/delete/delete_success
input.txt
expected_terminal.txt

login Enter transaction:
QuillFox9 Enter username:
delete Login successful.
Nimbus2 Enter transaction:
logout Enter username to delete:
User deleted.
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt

02 Nimbus2 FS 000005.00
00 QuillFox9 AA 000500.00

tests/general/invalid_empty_line
input.txt expected_terminal.txt

login Enter transaction:
QuillFox9 Enter username:
Login successful.
logout Enter transaction:
ERROR: invalid transaction
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt

00 QuillFox9 AA 000500.00

tests/general/invalid_unknown_wor
d input.txt expected_terminal.txt

login Enter transaction:
QuillFox9 Enter username:
frobnicate Login successful.
logout Enter transaction:
ERROR: invalid transaction
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt

00 QuillFox9 AA 000500.00

tests/general/invalid_wrong_case
input.txt expected_terminal.txt

login Enter transaction:
```

```
QuillFox9 Enter username:
Login Login successful.
logout Enter transaction:
ERROR: invalid transaction
Enter transaction:
Logout successful.
Enter transaction:
expected_daily.txt
00 QuillFox9 AA 000500.00
tests/general/permission_buy_as_sellstandar
d input.txt expected_terminal.txt

login Enter transaction:
CraterJax Enter username:
buy Login successful.
logout Enter transaction:
ERROR: not allowed for this account
type Enter transaction:
Logout successful.
Enter transaction:
expected_daily.txt
00 CraterJax SS 000050.00
tests/general/permission_sell_as_buystandar
d input.txt expected_terminal.txt

login Enter transaction:
Thistledown7 Enter username:
sell Login successful.
logout Enter transaction:
ERROR: not allowed for this account
type Enter transaction:
Logout successful.
Enter transaction:
expected_daily.txt
00 Thistledown7 BS 000100.00
tests/list/games_for_sale
input.txt expected_terminal.txt

login Enter transaction:
QuillFox9 Enter username:
list Login successful.
logout Enter transaction:
Games for sale:
Starlit Driftwood | MoonlitKai | 15.00
Emberfall Circuit | CraterJax | 30.00
Glass Orchard | MoonlitKai | 25.50
Quiet Nova Protocol | SilverQuasar1 | 100.00
Paperclip Expedition | QuillFox9 | 10.00
Driftwood Signal Nine | ShadowLynxRider | 50.00
Marrow Tide | CraterJax | 5.00
Enter transaction:
Logout successful.
Enter transaction:
expected_daily.txt
00 QuillFox9 AA 000500.00
tests/list/nothing_for_sale
games.txt differs from the standard fixture and is printed
below. input.txt expected_terminal.txt

login Enter transaction:
QuillFox9 Enter username:
list Login successful.
logout Enter transaction:
No games are currently for sale.
Enter transaction:
Logout successful.
Enter transaction:
expected_daily.txt
00 QuillFox9 AA 000500.00
games.txt
END
tests/list/read_only
```

input.txt expected\_terminal.txt

login Enter transaction:

QuillFox9 Enter username:

list Login successful.

list Enter transaction:

logout Games for sale:

Starlit Driftwood | MoonlitKai | 15.00

Emberfall Circuit | CraterJax | 30.00

Glass Orchard | MoonlitKai | 25.50

Quiet Nova Protocol | SilverQuasar1 | 100.00

Paperclip Expedition | QuillFox9 | 10.00

Driftwood Signal Nine | ShadowLynxRider | 50.00

Marrow Tide | CraterJax | 5.00

Enter transaction:

Games for sale:

Starlit Driftwood | MoonlitKai | 15.00

Emberfall Circuit | CraterJax | 30.00

Glass Orchard | MoonlitKai | 25.50

Quiet Nova Protocol | SilverQuasar1 | 100.00

Paperclip Expedition | QuillFox9 | 10.00

Driftwood Signal Nine | ShadowLynxRider | 50.00

Marrow Tide | CraterJax | 5.00

Enter transaction:

Logout successful.

Enter transaction:

expected\_daily.txt

00 QuillFox9 AA 000500.00

tests/list/reflects\_new\_listing

input.txt expected\_terminal.txt

login Enter transaction:

MoonlitKai Enter username:

**sell Login successful.**

Hollow Paper Kite Enter transaction:

20.00 Enter game name:

list Enter price:

logout Game put up for sale.

Enter transaction:

Games for sale:

Starlit Driftwood | MoonlitKai | 15.00

Emberfall Circuit | CraterJax | 30.00

Glass Orchard | MoonlitKai | 25.50

Quiet Nova Protocol | SilverQuasar1 | 100.00

Paperclip Expedition | QuillFox9 | 10.00

Driftwood Signal Nine | ShadowLynxRider | 50.00

Marrow Tide | CraterJax | 5.00

Hollow Paper Kite | MoonlitKai | 20.00

Enter transaction:

Logout successful.

Enter transaction:

expected\_daily.txt

03 Hollow Paper Kite MoonlitKai 020.00

00 MoonlitKai FS 000200.00

tests/list/stays\_after\_purchase

input.txt expected\_terminal.txt

login Enter transaction:

Thistledown7 Enter username:

buy Login successful.

Starlit Driftwood Enter transaction:

MoonlitKai Enter game name:

list Enter seller username:

logout Game purchased.

Enter transaction:

Games for sale:

Starlit Driftwood | MoonlitKai | 15.00

Emberfall Circuit | CraterJax | 30.00

Glass Orchard | MoonlitKai | 25.50

Quiet Nova Protocol | SilverQuasar1 | 100.00

Paperclip Expedition | QuillFox9 | 10.00

Driftwood Signal Nine | ShadowLynxRider | 50.00

Marrow Tide | CraterJax | 5.00

Enter transaction:

Logout successful.

Enter transaction:

```
expected_daily.txt
04 Starlit Driftwood MoonlitKai Thistledown7 015.00 00
Thistledown7 BS 000085.00

tests/login/login_case_mismatch
input.txt expected_terminal.txt

login Enter transaction:
quillfox9 Enter username:
logout ERROR: no such user
Enter transaction:
ERROR: please login first
Enter transaction:

expected_daily.txt: absent. No daily transaction file is expected.

tests/login/login_empty_username
input.txt expected_terminal.txt

login Enter transaction:
Enter username:
logout ERROR: invalid username
Enter transaction:
ERROR: please login first
Enter transaction:

expected_daily.txt: absent. No daily transaction file is expected.

tests/login/login_end_username
input.txt expected_terminal.txt

login Enter transaction:
END Enter username:
logout ERROR: invalid username
Enter transaction:
ERROR: please login first
Enter transaction:

expected_daily.txt: absent. No daily transaction file is expected.

tests/login/login_second_login_refused
input.txt expected_terminal.txt

login Enter transaction:
QuillFox9 Enter username:
login Login successful.
logout Enter transaction:
ERROR: already logged in
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt
00 QuillFox9 AA 000500.00

tests/login/login_success_admin
input.txt expected_terminal.txt

login Enter transaction:
QuillFox9 Enter username:
logout Login successful.
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt
00 QuillFox9 AA 000500.00

tests/login/login_success_standard
input.txt expected_terminal.txt

login Enter transaction:
MoonlitKai Enter username:
logout Login successful.
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt
00 MoonlitKai FS 000200.00

tests/login/login_unknown_user
input.txt expected_terminal.txt

login Enter transaction:
GhostAccount99 Enter username:
```

```
logout ERROR: no such user
Enter transaction:
ERROR: please login first
Enter transaction:

expected_daily.txt: absent. No daily transaction file is expected.

tests/login/transactions_before_login
input.txt expected_terminal.txt

create Enter transaction:
delete ERROR: please login first
logout Enter transaction:
login ERROR: please login first
QuillFox9 Enter transaction:
logout ERROR: please login first
Enter transaction:
Enter username:
Login successful.
Enter transaction:
Logout successful.
Enter transaction:
expected_daily.txt

00 QuillFox9 AA 000500.00

tests/logout/logout_after_logout
input.txt expected_terminal.txt

login Enter transaction:
QuillFox9 Enter username:
logout Login successful.
logout Enter transaction:
Logout successful.
Enter transaction:
ERROR: please login first
Enter transaction:

expected_daily.txt

00 QuillFox9 AA 000500.00

tests/logout/logout_blocks_until_login
input.txt expected_terminal.txt

login Enter transaction:
QuillFox9 Enter username:
logout Login successful.
create Enter transaction:
login Logout successful.
MoonlitKai Enter transaction:
logout ERROR: please login first
Enter transaction:
Enter username:
Login successful.
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt

00 MoonlitKai FS 000200.00

tests/logout/logout_not_logged_in
input.txt expected_terminal.txt

logout Enter transaction:
ERROR: please login first
Enter transaction:

expected_daily.txt: absent. No daily transaction file is expected.

tests/refund/buyer_over_max_credit
input.txt expected_terminal.txt

login Enter transaction:
QuillFox9 Enter username:
refund Login successful.
SilverQuasar1 Enter transaction:
MoonlitKai Enter buyer username:
100.00 Enter seller username:
logout Enter amount of credit:
ERROR: credit limit exceeded
Enter transaction:
Logout successful.
Enter transaction:
```

```
expected_daily.txt
00 QuillFox9 AA 000500.00
tests/refund/invalid_amount
input.txt expected_terminal.txt

login Enter transaction:
QuillFox9 Enter username:
refund Login successful.
Thistledown7 Enter transaction:
MoonlitKai Enter buyer username:
-10 Enter seller username:
refund Enter amount of credit:
Thistledown7 ERROR: invalid amount
MoonlitKai Enter transaction:
0.00 Enter buyer username:
refund Enter seller username:
Thistledown7 Enter amount of credit:
MoonlitKai ERROR: invalid amount
abc Enter transaction:
logout Enter buyer username:
Enter seller username:
Enter amount of credit:
ERROR: invalid amount
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt
00 QuillFox9 AA 000500.00
tests/refund/refund_as_nonadmin input.txt
expected_terminal.txt

login Enter transaction:
MoonlitKai Enter username:
refund Login successful.
logout Enter transaction:
ERROR: not allowed for this account
type Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt
00 MoonlitKai FS 000200.00
tests/refund/same_person_twice
input.txt expected_terminal.txt

login Enter transaction:
QuillFox9 Enter username:
refund Login successful.
MoonlitKai Enter transaction:
MoonlitKai Enter buyer username:
logout Enter seller username:
ERROR: buyer and seller must be
different Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt
00 QuillFox9 AA 000500.00
tests/refund/seller_not_enough_credit input.txt
expected_terminal.txt

login Enter transaction:
QuillFox9 Enter username:
refund Login successful.
Thistledown7 Enter transaction:
CraterJax Enter buyer
username: 100.00 Enter seller
username:

logout Enter amount of credit:
ERROR: not enough credit
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt
```

```
00 QuillFox9 AA 000500.00

tests/refund/successful_refund
input.txt expected_terminal.txt

login Enter transaction:
QuillFox9 Enter username:
refund Login successful.
Thistledown7 Enter transaction:
MoonlitKai Enter buyer username:
20.00 Enter seller username:
logout Enter amount of credit:
Refund complete.
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt

05 Thistledown7 MoonlitKai 000020.00
00 QuillFox9 AA 000500.00

tests/refund/unknown_person
input.txt

expected_terminal.txt login
Enter transaction:

QuillFox9 Enter
username: refund Login
successful.
Ghostwalker12 Enter
transaction: logout Enter buyer
username:
ERROR: no such user
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt

00 QuillFox9 AA 000500.00

tests/refund/unknown_seller
input.txt expected_terminal.txt

login Enter transaction:
QuillFox9 Enter username:
refund Login successful.
Thistledown7 Enter transaction:
Ghostwalker12 Enter buyer
username: logout Enter seller
username:
ERROR: no such user
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt

00 QuillFox9 AA 000500.00

tests/sell/admin_can_sell
input.txt expected_terminal.txt

login Enter transaction:
QuillFox9 Enter username:
sell Login successful.

Admin Listed Game Enter
transaction: 8.00 Enter game name:
logout Enter price:
Game put up for sale.
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt

03 Admin Listed Game QuillFox9 008.00
00 QuillFox9 AA 000500.00

tests/sell/invalid_price
input.txt expected_terminal.txt

login Enter transaction:
MoonlitKai Enter username:
```

```
sell Login successful.
Hollow Paper Kite Enter
transaction: abc Enter game name:
logout Enter price:
ERROR: invalid price
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt

00 MoonlitKai FS 000200.00

tests/sell/name_END
input.txt expected_terminal.txt

login Enter transaction:
MoonlitKai Enter username:
sell Login successful.
END Enter transaction:
logout Enter game name:
ERROR: invalid game name
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt

00 MoonlitKai FS 000200.00
tests/sell/name_already_listed
input.txt expected_terminal.txt

login Enter transaction:
CraterJax Enter username:
sell Login successful.
Starlit Driftwood Enter transaction:
9.99 Enter game name:
logout ERROR: game already for sale
Enter transaction:
ERROR: invalid transaction
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt

00 CraterJax SS 000050.00

tests/sell/name_exactly_25
input.txt expected_terminal.txt

login Enter transaction:

MoonlitKai Enter username:
sell Login successful.
ABCDEFGHJKLMNOPQRSTUVWXYZ Enter
transaction: 12.00 Enter game name:
logout Enter price:
Game put up for sale.
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt

03 ABCDEFGHJKLMNOPQRSTUVWXYZ MoonlitKai 012.00 00
MoonlitKai FS 000200.00

tests/sell/name_too_long
input.txt expected_terminal.txt

login Enter transaction:
MoonlitKai Enter username:
sell Login successful.
ABCDEFGHIJKLMNOPQRSTUVWXYZ Enter
transaction: logout Enter game name:
ERROR: game name too long (max 25)
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt

00 MoonlitKai FS 000200.00

tests/sell/price_at_max
input.txt expected_terminal.txt

login Enter transaction:
```

```
MoonlitKai Enter username:
sell Login successful.
Hollow Paper Kite Enter transaction:
999.99 Enter game name:
logout Enter price:
Game put up for sale.
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt

03 Hollow Paper Kite MoonlitKai 999.99
00 MoonlitKai FS 000200.00

tests/sell/price_over_max
input.txt expected_terminal.txt

login Enter transaction:
MoonlitKai Enter username:
sell Login successful.
Hollow Paper Kite Enter transaction:
1000.00 Enter game name:
logout Enter price:
ERROR: price too high (max 999.99)
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt

00 MoonlitKai FS 000200.00

tests/sell/sellstandard_can_sell
input.txt expected_terminal.txt

login Enter transaction:
CraterJax Enter username:
sell Login successful.
New Sell Standard Listing Enter transaction:
7.00 Enter game name:
logout Enter price:
Game put up for sale.
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt

03 New Sell Standard Listing CraterJax 007.00
00 CraterJax SS 000050.00

tests/sell/successful_sell
input.txt expected_terminal.txt

login Enter transaction:
MoonlitKai Enter username:
sell Login successful.
Hollow Paper Kite Enter transaction:
20.00 Enter game name:
logout Enter price:
Game put up for sale.
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt

03 Hollow Paper Kite MoonlitKai 020.00
00 MoonlitKai FS 000200.00

tests/sell/whole_dollar_price
input.txt expected_terminal.txt

login Enter transaction:
MoonlitKai Enter username:
sell Login successful.
Round Number Game Enter transaction:
15 Enter game name:
logout Enter price:
Game put up for sale.
Enter transaction:
Logout successful.
Enter transaction:

expected_daily.txt

03 Round Number Game MoonlitKai 015.00
```

## 5. test plan

### organization

Each test has its own folder under tests/<transaction>/<test-name>/. Normal tests have input.txt, accounts.txt, games.txt and expected\_terminal.txt. We include expected\_daily.txt when a daily file is expected. Bad-file tests leave out a file or use a damaged record on purpose.

### work split

victor wrote the login, logout, create, delete and general/permission tests (31). arian wrote sell, buy, refund, addcredit, list and bad-file tests (43).

### running the tests

When the Front End is available, run from phase1-frontend-tests/: FRONTEND=/absolute/path/to/frontend ./scripts/run\_all.sh. To run one test: ./scripts/run\_test.sh tests/sell/successful\_sell results/manual. The

program is called as:

**frontend accounts.txt games.txt daily.txt < input.txt > terminal.txt.**

### checking output

The runner uses diff to compare terminal and daily output with the expected files. If expected\_daily.txt is missing, it checks that no daily file was created. it uses a 10-second limit when timeout or gtimeout is installed. without either tool, there is no automatic time limit. phase 1 does not include results from running a Front End.

### saving results

Each run has a results/run\_<date>\_<time>/ folder. summary.txt lists PASS or FAIL, and report.txt gives totals. the details folders keep the actual output, differences, stderr and exit code. folder names include the group and test name so tests with the same name do not overwrite each other. results/latest points to the newest run.

### comparing later runs

compare\_runs.sh compares two summaries by group and test name. check\_formats.sh checks record lengths and skips the deliberately corrupt fixtures. if a client answer changes a decision, we update the notes and affected tests together.

### directory structure

#### directory structure

```
phase1-frontend-tests/
README.md
HANDIN.md
requirements_decisions.md
test_plan.md
scripts/
check_formats.sh
compare_runs.sh
run_all.sh
run_test.sh
results/
README.md
tests/
addcredit/
admin_adds_to_other/
exactly_session_limit/
invalid_amount/
over_max_credit/
over_session_limit/
standard_adds_own/
unknown_user/
bad_files/
corrupt_accounts_file/
corrupt_games_file/
missing_accounts_file/
missing_games_file/
buy/
admin_can_buy/
already_owned/
buying_own_game/
game_not_found/
new_listing_same_session/
not_enough_credit/
seller_over_max_credit/
successful_buy/
```

```
create/  
create_as_nonadmin/  
create_duplicate_deleted_this_session/  
create_duplicate_existing/  
create_empty_username/  
create_end_username/  
create_success/  
create_success_15char/  
create_too_long/  
  
invalid_user_type/
```

#### **directory structure (continued)**

```
delete/  
delete_as_nonadmin/  
delete_empty_username/  
delete_end_username/  
delete_nonexistent/  
delete_self_admin/
```

```
delete_success/  
general/  
invalid_empty_line/  
invalid_unknown_word/  
invalid_wrong_case/  
permission_buy_as_sellstandard/  
permission_sell_as_buystandard/  
list/  
games_for_sale/  
nothing_for_sale/  
read_only/  
reflects_new_listing/  
stays_after_purchase/  
login/  
login_case_mismatch/  
login_empty_username/  
login_end_username/  
login_second_login_refused/  
login_success_admin/  
login_success_standard/  
login_unknown_user/  
transactions_before_login/  
logout/  
logout_after_logout/  
logout_blocks_until_login/  
logout_not_logged_in/  
refund/  
buyer_over_max_credit/  
invalid_amount/  
refund_as_nonadmin/  
same_person_twice/  
seller_not_enough_credit/  
successful_refund/  
unknown_person/  
unknown_seller/  
sell/  
admin_can_sell/  
invalid_price/  
name_END/
```

#### **directory structure (continued)**

```
name_already_listed/  
name_exactly_25/  
name_too_long/  
price_at_max/  
price_over_max/  
sellstandard_can_sell/  
successful_sell/  
whole_dollar_price/
```

#### **6. script printouts**

```
scripts/check_formats.sh  
check_formats.sh
```

```
#!/bin/bash  
# checks the fixed-width files in the test suite  
# accounts.txt every line 28 characters  
# games.txt every line 48 characters  
# expected_daily.txt line length depends on the transaction code  
# 00/01/02/06 = 31, 03 = 51, 04 = 67, 05 = 44
```

```

# skip the intentionally broken fixture folders
# usage: ./scripts/check_formats.sh

SUITE="$ (cd "$(dirname "$0")/.." && pwd) "
BAD=0

check_length() {
    awk -v want="$2" 'length($0) != want { printf "%s line %d has %d characters, expected %d\n", FILENAME,
    FNR, length($0), want; bad = 1 }
    END { exit bad }' "$1"
}

while IFS= read -r FILE; do
    case "$FILE" in *corrupt*) continue ;; esac
    check_length "$FILE" 28 || BAD=1
done < <(find "$SUITE/tests" -name accounts.txt | sort)

while IFS= read -r FILE; do
    case "$FILE" in *corrupt*) continue ;; esac
    check_length "$FILE" 48 || BAD=1
done < <(find "$SUITE/tests" -name games.txt | sort)

while IFS= read -r FILE; do
    awk '
    { code = substr($0, 1, 2) }
    code == "00" || code == "01" || code == "02" || code == "06" { want = 31 }
    code == "03" { want = 51 }
    code == "04" { want = 67 }
    code == "05" { want = 44 }
    code !~ /^0[0-6]$/ { printf "%s line %d has an unknown code\n", FILENAME, FNR; bad = 1; next }
    length($0) != want { printf "%s line %d has %d characters, expected %d\n", FILENAME, FNR, length($0), want; bad = 1 }
    END { exit bad }
    ' "$FILE" || BAD=1
    LAST="$(tail -n 1 "$FILE" | cut -c1-2)"
    if [ "$LAST" != "00" ]; then
        echo "$FILE does not end with a 00 line"
        BAD=1
    fi
done < <(find "$SUITE/tests" -name expected_daily.txt | sort)

if [ "$BAD" -eq 0 ]; then
    echo "All test files have the correct line lengths."
    exit 0
fi
echo "Some test files have the wrong format."
exit 1

scripts/compare_runs.sh
compare_runs.sh

#!/bin/bash
# compare two saved test runs
# usage: ./scripts/compare_runs.sh results/run_<old> results/run_<new>
# useful after changing the Front End

OLD="$1"
NEW="$2"

if [ ! -f "$OLD/summary.txt" ] || [ ! -f "$NEW/summary.txt" ]; then
    echo "usage: compare_runs.sh <old_run_directory> <new_run_directory>"
    exit 2
fi

awk '
NR == FNR { old[$2/"$3] = $1; next }
{
    before = old[$2/"$3]
    if (before == "") before = "MISSING"
    if (before != $1) {
        label = "CHANGED"

        if (before == "PASS" && $1 == "FAIL") label = "REGRESSION"
        if (before == "FAIL" && $1 == "PASS") label = "FIXED"
        if (before == "MISSING") label = "NEW TEST"
        printf "%-11s %s (%s -> %s)\n", label, $3, before, $1
        changed++
    }
}
END { if (changed == 0) print "No differences between the two runs." }
' "$OLD/summary.txt" "$NEW/summary.txt"

echo ""
echo "Old: $(grep -c '^PASS' "$OLD/summary.txt") passed, $(grep -c '^FAIL' "$OLD/summary.txt")
failed" echo "New: $(grep -c '^PASS' "$NEW/summary.txt") passed, $(grep -c '^FAIL'
"$NEW/summary.txt") failed"

```

```

scripts/run_all.sh
run_all.sh

#!/bin/bash
# Runs the Phase 1 test suite and saves the results for this run.
# Make sure the scripts are executable before running them.

# run all tests and save this run in a new results folder
# usage: ./scripts/run_all.sh [text]
# optional argument filters by folder name
# (for example ./scripts/run_all.sh sell runs the sell tests only)

SUITE="$(cd "$(dirname "$0")/.." && pwd)"
FILTER="$1"
STAMP="$(date +%Y%m%d_%H%M%S)"
RUN="$SUITE/results/run_${STAMP}"
mkdir -p "$RUN/details"
SUMMARY="$RUN/summary.txt"
: > "$SUMMARY"

find "$SUITE/tests" -mindepth 2 -maxdepth 2 -type d | sort > "$RUN/test_list.txt"

TOTAL=0
PASSED=0
FAILED=0

while IFS= read -r DIR; do
if [ -n "$FILTER" ]; then
case "$DIR" in
*"$FILTER"*) ;;
*) continue ;;
esac
fi
NAME="$(basename "$DIR")"
GROUP="$(basename "$(dirname "$DIR")")"
if "$SUITE/scripts/run_test.sh" "$DIR" "$RUN/details" > /dev/null;
then echo "PASS $GROUP $NAME" >> "$SUMMARY"
PASSED=$((PASSED + 1))
else
echo "FAIL $GROUP $NAME" >> "$SUMMARY"
FAILED=$((FAILED + 1))
fi
TOTAL=$((TOTAL + 1))
done < "$RUN/test_list.txt"

rm -f "$RUN/test_list.txt"

{
echo "Run: run_${STAMP}"
echo "Front End: ${FRONTEND:-./frontend}"
echo "Total: $TOTAL Passed: $PASSED Failed: $FAILED"
echo ""
echo "Per group (passed / total):"
awk '{ total[$2]++; if ($1 == "PASS") pass[$2]++ }
END { for (g in total) printf " %-20s %d / %d\n", g, pass[g], total[g] }' "$SUMMARY" |
sort echo ""
echo "Failed tests:"

grep '^FAIL' "$SUMMARY" | awk '{ print " " $3 }'

scripts/run_all.sh (continued)
run_all.sh

} > "$RUN/report.txt"

cat "$RUN/report.txt"

rm -f "$SUITE/results/latest"
ln -s "run_${STAMP}" "$SUITE/results/latest"

if [ "$FAILED" -gt 0 ]; then
exit 1
fi
exit 0

scripts/run_test.sh
run_test.sh

#!/bin/bash
# run one test and compare the output files
# usage: ./scripts/run_test.sh <test_directory> <results_directory>
# FRONTEND can point to the built program, otherwise it uses ./frontend

TEST_DIR="$1"
OUT_ROOT="$2"
FRONTEND="${FRONTEND:-./frontend}"

```

```

if [ -z "$TEST_DIR" ] || [ -z "$OUT_ROOT" ]; then
echo "usage: run_test.sh <test_directory> <results_directory>"
exit 2
fi

case "$FRONTEND" in
/*) ;;
*) FRONTEND="$(pwd)/$FRONTEND" ;;
esac

TEST_DIR="$(cd "$TEST_DIR" && pwd)"
mkdir -p "$OUT_ROOT"
OUT_ROOT="$(cd "$OUT_ROOT" && pwd)"
GROUP="$(basename "$(dirname "$TEST_DIR")")"
NAME="${GROUP}_${basename "$TEST_DIR"}"
RES="$OUT_ROOT/$NAME"
mkdir -p "$RES"
rm -f "$RES"/*

if command -v timeout >/dev/null 2>&1; then
LIMIT="timeout 10"
elif command -v gtimeout >/dev/null 2>&1; then
LIMIT="gtimeout 10"
else
LIMIT=""
fi

cd "$TEST_DIR"
$LIMIT "$FRONTEND" accounts.txt games.txt "$RES/daily.txt" < input.txt > "$RES/terminal.txt"
2> "$RES/stderr.txt"
CODE=$?
echo "$CODE" > "$RES/exit_code.txt"

STATUS=PASS

if [ "$CODE" -ge 124 ]; then
STATUS=FAIL
echo "program crashed, was not found, or timed out (exit code $CODE)" >
"$RES/note.txt" fi

if ! diff expected_terminal.txt "$RES/terminal.txt" > "$RES/terminal.diff" 2>&1;
then STATUS=FAIL

fi

if [ -f expected_daily.txt ]; then
if ! diff expected_daily.txt "$RES/daily.txt" > "$RES/daily.diff" 2>&1;
then STATUS=FAIL

scripts/run_test.sh (continued)
run_test.sh

fi
else
if [ -f "$RES/daily.txt" ]; then
echo "a daily transaction file was written but none was expected" >
"$RES/daily.diff" STATUS=FAIL
fi
fi

echo "$STATUS" > "$RES/status.txt"
echo "$STATUS $NAME"

if [ "$STATUS" = "PASS" ]; then
exit 0
fi
exit 1

```